

Ongoing Work on Contextual Neural Machine Translation

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Motivation

State of the Art Literature

Ongoing Work at i6

Experiments

Summary and Future Work

- ▶ Most neural machine translation (NMT) systems translate sentence by sentence
- ▶ Make use of other sentences in addition to exploit context information
 - ▷ reduce ambiguities in translation choices
 - ▷ example: cross-sentence anaphora [Voita & Sennrich⁺ 18]

An English→German example of a cross-sentence anaphora problem)p. The translation of the pronoun 'it' depends on the previous sentence.

Source sentences	Correct translation of 'it'
<i>The monkey ate the banana. Because it was hungry.</i>	er
<i>The monkey ate the banana. Because it ripe.</i>	sie
<i>The monkey ate the banana. Because it was tea-time.</i>	es

- ▶ **Single encoder: concatenate sentences on source-/target side**
[Tiedemann & Scherrer 17, Agrawal & Turchi⁺ 18]

- ▶ **Multi-Encoder with attention mechanism**
 - ▷ **Local context: feed surrounding sentences into separate encoders. Combine context vectors:**
 - concatenation (like in [Zoph & Knight 16])
 - gated sum [Bawden & Sennrich⁺ 18]
 - hierarchical attention [Bawden & Sennrich⁺ 18]

Remark: above methods only consider *local* context

1. Reproduce results from [Agrawal & Turchi⁺ 18]

- ▶ using Transformer model as well
- ▶ first, find good baseline: hyperparameter search
 - ▷ weight-tying
 - ▷ learning rates
 - ▷ checkpoints
 - ▷ ...
- ▶ concatenate sentences with special token <BREAK>

Thank you for your attention

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